

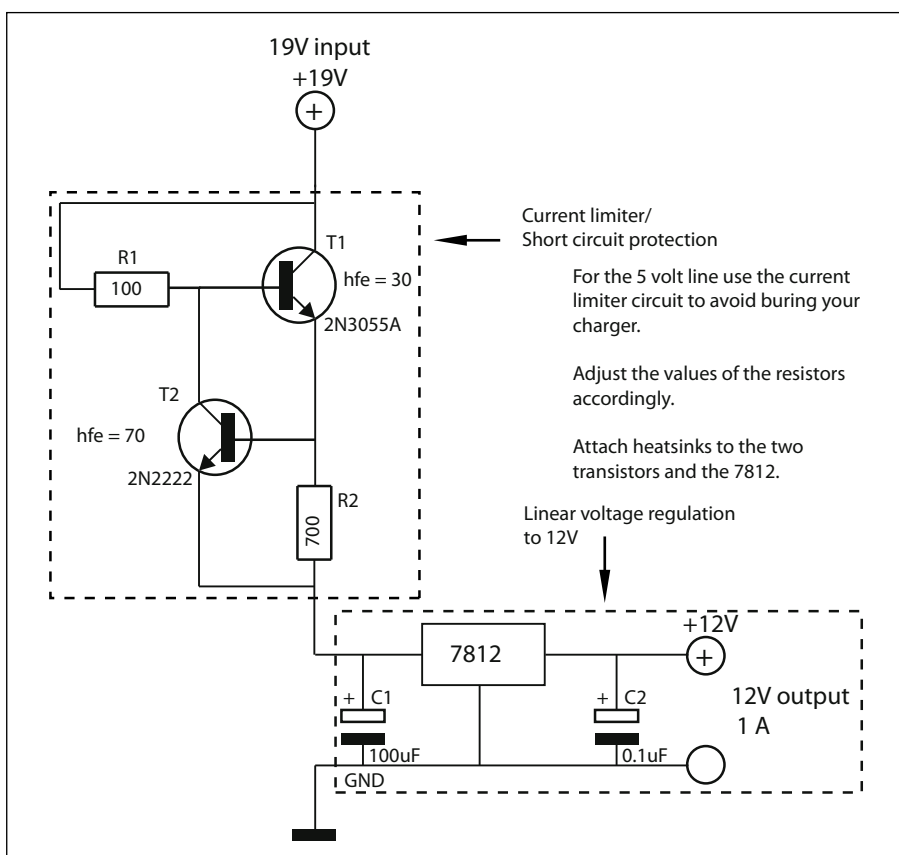
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Logitech G400

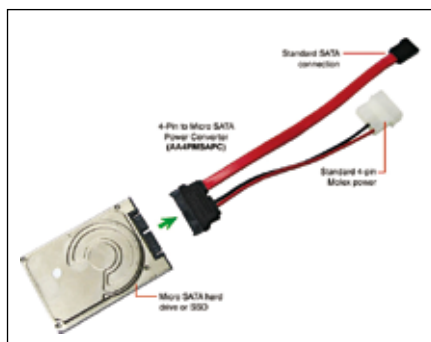
Logitech introduces rock-solid performer optical gaming mouse G400, at ₹2,095. <http://dgit.in/P4gfUd>



Circuit for the current limiter + voltage regulator



12 V on the yellow wire and 5 V on the red



mSATA power and data cable

an input of 1 ampere and have a heatsink attached to it as it will get quite depending upon the power dissipated across it.

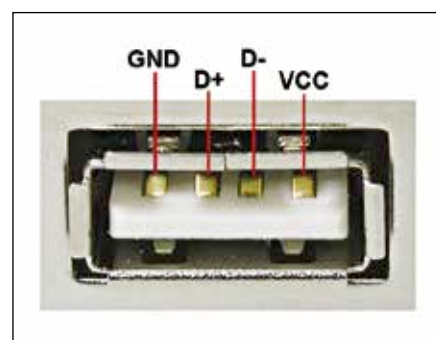
Similarly build another circuit for the 5 volt line, use the simulator to help you out. Cross reference the transistor data sheets to get the proper parameters.

Usage scenarios

1. If you are using the adapter on a simple 2.5-inch HDD or SSD then you can directly connect the adapter to a USB cable and then your adapter to your

HDD. For 1.8 -inch drives, you'll need to further reduce the voltage down to 3.3 volts and modify the wiring as they do not use the same SATA power connectors like the others do.

2. When using this on a 3.5 -inch HDD, you'll need to make use of the above circuit as these drives are heavily dependent on the 12 volt input line. Take a 4-pin Molex to SATA power adapter like the one below and snip off the molex connector. The yellow wire is where the 12 volt output from the



You'll need the VCC and GND to get 5 Volts

voltage regulator goes in. The red wire is the 5 volt line which is sourced from the USB port.

A Y cable needs to be used for this purpose and ensure that the two male USB A ports are inserted into either a powered hub or ports which are connected to different headers. This is because each header is limited to 500mA and your drive will need nearly 1A or a bit more for the initial spin-up stage. For this the power needs to be sourced from two different headers on the motherboard. If you are using a laptop then insert them into ports at opposite sides of the laptop as they are from different headers. If you only have ports which are close together then most probably they obtain power from the same header. In case of powered hubs you need not worry as each port is capable of 500mA.

Always have both inputs (from the USB ports and the laptop charger) connected to individual current limiters before connecting them to the SATA power adapter. This is because your drive will try to draw as much as power that it requires and when this amount is greater than what your motherboard can provide, it'll blow a fuse. Or worse, the VRM circuitry on your laptop/desktop's motherboard.

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